

Learnable Lemma Embeddings for GLP Models

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Abstract

This paper introduces a learnable lemma embedding layer for Grammar Lemma Paged Language Models (GLP), replacing the earlier bit-vector and tri-hot encoding with a continuous, fixed-dimensional geometric space. The new layer enables lemma selection through cosine-similarity scoring within contextually activated pages, while preserving GLP’s deterministic grammar structure, paging organisation, and dual-channel generative loop. Lemma embeddings are trained through a two-stage mechanism: the lemma MLP predicts a target embedding optimised via cosine similarity, and after each batch the stored lemma embedding is updated through an α -damped mixing rule.

Empirical tests on a very small 9 MB GLP model trained on a micro-world demonstrate the structural role of the learnable embedding space. With $\alpha = 0$ (no learning), the model collapses into a small set of attractors and exhibits limited variation. With low learning rates, the embedding space absorbs world structure but becomes unstable without damping, leading to semantic drift and degraded page coherence. With higher learning rates, the model rapidly acquires world structure, and α -damping ensures convergence and stable long-horizon behaviour. Across all conditions, pure sampling remains fast and expressive, and the learned embedding space consistently outperforms fixed bit-based encodings such as tri-hot.

A larger 33 MB GLP model trained on *Thuvia, Maid of Mars* (Gutenberg) further illustrates the practical effectiveness of the approach: the learned lemma space produces coherent pulp-fiction narratives that can be remastered by transformer models (Mistral 7B and a larger LLM), resulting in the full volume *The Cosmic Parrot: Triadic Pulp Fiction*. These results show that a lightweight structural language model equipped with learnable lemma embeddings can serve as a fast, stable, and domain-faithful generative engine within hybrid LLM+GLP systems.

1 Introduction

Grammar Lemma Paged Language Models (GLP) constitute a structurally governed class of next-token generators built around deterministic grammar structure, paged lemma vocabularies, and fixed-dimensional output geometry. In contrast to transformer-based language models, which unify grammar, lexical content, and context into a single high-dimensional embedding space, GLP decomposes generation into two alternating channels: a compact grammar model that determines structural cadence, and a lemma model whose vocabulary is organised into contextually activated pages. This dual-channel architecture provides explicit structural control, domain isolation, and stable long-horizon behaviour, making GLP a modular alternative to monolithic embedding-softmax systems.

In its original formulation, GLP represented lemma tokens through hybrid bit-vectors and tri-hot codes. While effective for fixed-geometry decoding, these discrete encodings limit semantic expressiveness, slow down sampling, and constrain vocabulary growth. This paper introduces a continuous, learnable lemma embedding layer that replaces the discrete codebook with a fixed-dimensional geometric space. Lemma selection is performed through cosine-similarity scoring within the active page, filtered by grammar compatibility, yielding fast and stable ranking behaviour. The embedding layer is trained

through a two-stage mechanism: the lemma MLP predicts a target embedding optimised via cosine similarity, and after each training batch the stored lemma embedding is updated through an α -damped mixing rule. Empirically, this damping mechanism is essential for long-term stability: without damping, extended training causes embedding drift, degraded page coherence, and slower generation; with damping, the embedding space converges and sampling remains fast even after millions of training steps.

The learnable embedding layer integrates seamlessly with GLP’s structural commitments. Grammar tokens remain deterministic symbols with strict structural filtering; pages remain discrete geometric regions; and lemma selection remains constrained by grammar compatibility and page activation. The result is a hybrid symbolic–geometric architecture: grammar provides deterministic structure, paging provides domain isolation, and lemma embeddings provide continuous semantic variation.

To evaluate the practical effectiveness of the approach, we trained a 33 MB GLP model on *Thuvia, Maid of Mars* (Gutenberg) and generated narrative sequences using the GLP narrator pipeline. The generated stories were subsequently remastered by transformer models (Mistral 7B and a larger LLM), producing the full volume *The Cosmic Parrot: Triadic Pulp Fiction*. This case study demonstrates that a lightweight structural model equipped with learnable lemma embeddings can serve as a fast, stable, and domain-faithful generative engine within hybrid GLP+LLM systems.

The remainder of this paper formalises the learnable lemma embedding layer, describes its integration into GLP’s dual-channel architecture, analyses the role of α -damping in embedding stability, and presents empirical results from narrator-pipeline generation and hybrid LLM remastering.

2 Learnable Lemma Embeddings

The original GLP architecture represented lemma tokens using hybrid bit-vectors and tri-hot codes. While this fixed-geometry encoding ensured deterministic output structure, it limited semantic expressiveness, constrained vocabulary growth, and imposed discrete matching during lemma selection. In this work, we replace the discrete lemma encoding with a continuous, learnable embedding space that integrates directly into GLP’s dual-channel architecture while preserving its structural commitments.

2.1 Continuous lemma representation

Each lemma token ℓ is assigned a fixed-dimensional embedding $e_\ell \in \mathbb{R}^d$, where the lemma dimension d is a global hyperparameter. Embeddings are deterministically initialised from the token’s hashed identifier, ensuring reproducibility and stable early training behaviour. Unlike transformer models, GLP does not use embeddings for grammar tokens; grammar remains symbolic and deterministic. The embedding space is used exclusively for lemma selection, providing a continuous geometric layer within GLP’s structurally governed generative loop.

2.2 Lemma MLP and embedding prediction

During lemma generation, the lemma MLP receives the encoded model input (context window, narrative memory, and story embedding) and produces two outputs:

1. a predicted lemma embedding $\hat{e} \in \mathbb{R}^d$,
2. a vector of page logits determining the most relevant lemma page.

The predicted embedding is normalised to unit length. Page logits are used to rank pages in descending order, enabling context-driven page activation with automatic fallback when the primary page yields no valid candidates.

2.3 Cosine-similarity lemma selection

Lemma selection is performed geometrically. For each candidate lemma ℓ in the active page, GLP computes the cosine similarity between the predicted embedding and the stored embedding:

$$\text{score}(\ell) = \cos(\hat{e}, e_\ell).$$

Grammar compatibility filtering removes lemma tokens that violate structural constraints learned during curriculum training. The remaining candidates are sorted by cosine similarity, and top- k sampling with temperature is applied to produce the final lemma token. This mechanism replaces discrete codebook matching with a continuous geometric ranking, yielding faster sampling and richer variation.

2.4 Two-stage embedding learning

Lemma embeddings are trained through a two-stage process:

1. **Cosine-similarity optimisation.** During backpropagation, the lemma MLP is trained to predict embeddings close to the stored lemma embeddings. The loss term

$$\mathcal{L}_{\text{emb}} = 1 - \cos(\hat{e}, e_\ell)$$

encourages alignment between predicted and stored embeddings.

2. **Post-batch α -damped update.** After each training batch, the stored embedding e_ℓ is updated by mixing it with the predicted embedding:

$$e_\ell \leftarrow (1 - \alpha)e_\ell + \alpha\hat{e},$$

where α decreases gradually from α_{max} to α_{min} over a fixed number of training steps. This damping mechanism allows rapid early learning while ensuring long-term stability.

2.5 Role of α -damping

Empirical tests show that α -damping is essential for stable training. Without damping, lemma embeddings continue to drift even after many epochs, causing page coherence to degrade, cosine-similarity ranking to fluctuate, and sampling to slow down. With damping, the embedding space converges smoothly, preserving page structure and maintaining fast inference. This demonstrates that the post-batch update rule is not merely an optimisation detail but a structural requirement for GLP’s continuous lemma space.

2.6 Integration with paging and grammar structure

The learnable lemma embedding layer preserves all structural components of GLP:

- **Grammar tokens** remain deterministic symbols with strict structural filtering.
- **Pages** remain discrete geometric regions that isolate domain-specific lemma clusters.
- **Compatibility masks** ensure that lemma selection respects canonical grammar.
- **Narrative memory** continues to guide page activation and long-horizon coherence.

The result is a hybrid symbolic-geometric architecture: grammar provides deterministic structure, paging provides domain isolation, and lemma embeddings provide continuous semantic variation.

2.7 Practical impact

The learnable lemma embedding layer significantly improves GLP’s practical performance. Sampling becomes faster, vocabulary growth becomes fully dynamic, and fewer epochs are required to obtain a stable model. The continuous lemma space also aligns more closely with the geometric behaviour of transformer models, enabling natural hybrid pipelines where GLP provides the world engine and LLMs provide stylistic refinement.

This section establishes the learnable lemma embedding layer as a foundational extension of GLP, enabling a continuous semantic space within a structurally governed generative architecture.

3 Advantages

The learnable lemma embedding layer introduces several practical and architectural advantages over the earlier bit–vector and tri–hot encoding used in GLP. These advantages arise from the combination of continuous lemma embeddings, cosine–similarity scoring, and the α –damped post–batch update rule.

Continuous geometric lemma space

Lemma tokens are represented in a fixed–dimensional continuous space, making the lemma representation much closer to the geometric behaviour of modern LLM output embeddings. This improves hybrid integration in narrator pipelines and enables richer semantic variation within GLP’s structurally governed architecture. GLP remains a structural language model: grammar is deterministic, pages are discrete, and generation follows the dual–channel GLP loop rather than transformer–style attention.

Dynamic vocabulary without geometric constraints

Because lemma embeddings live in a fixed–dimensional vector space, vocabulary growth is decoupled from model geometry. New lemma tokens can be added dynamically without altering the output dimension of the lemma MLP or the paging structure. This removes the structural limitations of tri–hot encoding, where vocabulary size is tied directly to output geometry.

Fast sampling

Cosine–similarity scoring provides a lightweight and stable ranking mechanism for lemma selection. Combined with page–restricted candidate sets and grammar compatibility filtering, sampling becomes significantly faster in practice.

Improved generation quality and richness

The continuous embedding space enables fine–grained semantic variation within pages, yielding richer lemma diversity and more expressive narrative output. Cosine–similarity scoring naturally clusters semantically related lemma tokens, improving stylistic variation without compromising canonical structure.

Fewer epochs required for stable training

The α –damped post–batch update rule allows lemma embeddings to converge rapidly during early training while stabilising over time. Empirically, models trained without damping show degraded performance after many epochs: embedding drift slows down generation and reduces page coherence. With damping, the embedding space converges smoothly, requiring fewer epochs to reach a stable and usable model state.

No observed disadvantages

Across all experiments, including narrator–pipeline generation and hybrid LLM remastering, no disadvantages were observed. The learnable lemma embedding layer integrates cleanly with GLP’s deterministic grammar structure, paging organisation, and PRE–based sentence validation, while improving speed, stability, and expressive power.

4 A Trainable World Engine

Grammar Lemma Paged Language Models (GLP) are not merely structural language models; they function as *trainable world engines*. When trained on a single book or corpus, GLP absorbs the underlying narrative geometry: its grammar cadence, lemma distribution, page structure, stylistic attractors, and domain–specific semantic regularities. Through curriculum training, paging dynamics, and the learnable lemma embedding layer introduced in this paper, GLP internalises the world model implicit in the source text and reconstructs it as a compact, inspectable generative substrate.

This world engine behaviour differs fundamentally from transformer–based LLMs. GLP does not memorise surface text or rely on high–dimensional embeddings; instead, it learns a structural ontology of the domain. Grammar tokens encode the syntactic and rhythmic backbone of the world, lemma pages capture domain–specific semantic clusters, and the continuous lemma embedding space provides fine–grained variation within these clusters. The result is a lightweight model that can reproduce the *world* of a book rather than its literal sentences.

Once trained, GLP can be used to generate *proto–stories*: structurally coherent, canon–faithful narrative sequences that reflect the learned world model. These proto–stories serve as raw material for hybrid refinement. In the narrator pipeline, GLP provides the world engine, sampling provides expressive exploration, and transformer models (such as Mistral 7B and a larger LLM) provide iterative stylistic polishing. This hybrid process mirrors human–LLM collaboration, where a human supplies world knowledge and structural constraints while the LLM supplies surface fluency. GLP is not equivalent to a human and does not replicate human creativity, but it provides a structurally governed world model that plays a similar functional role in the generative pipeline.

This approach was demonstrated empirically by training a 33 MB GLP model on *Thuvia, Maid of Mars* (Gutenberg). The trained model acted as a compact world engine for Barsoom–style pulp fiction, generating proto–stories that were subsequently refined by transformer models into the full volume *The Cosmic Parrot: Triadic Pulp Fiction*. This case study illustrates how GLP can absorb a narrative domain, generate new stories within that domain, and collaborate with LLMs to produce polished literary output.

GLP therefore occupies a unique position in the Triadic Cosmos ecosystem: a modular, trainable world engine that provides structural, canonical, and domain–faithful generative behaviour, enabling hybrid pipelines where lightweight structural models and large transformers complement each other in the creation of new narrative worlds.

5 Narrative Behaviour Under Hybrid GLP+LLM Generation

The narrative corpus analysed in this section was produced by the narrator pipeline using the 33 MB GLP world model trained on *Thuvia, Maid of Mars*, followed by a light transformer–based remastering step using Mistral 7B. This remastering was intentionally minimal: the LLM received only high–level narrative guidance such as

Make sure there is proper and exciting narrative arc. Keep all story elements related and realistic. Fill the gaps. A male main character is called Marvin and a female main character is called Alice.

No stylistic constraints, world–building hints, or Barsoom references were provided. This setup allows us to observe how GLP’s structural world model interacts with a transformer LLM that performs

interpretive refinement rather than content invention. The resulting stories therefore reflect a hybrid generative process: GLP supplies the world geometry and proto-story structure, while the LLM supplies surface coherence, narrative arc formation, and stylistic polish.

5.1 GLP as the structural substrate

Even after LLM remastering, the underlying GLP world model remains clearly visible. Across all generated stories, GLP consistently provides:

- Barsoom-like spatial geometry (palaces, towers, deserts, barrooms),
- a stable social ontology (Pathans, nobles, warriors, princesses),
- action grammar (pursuit, confrontation, rescue, infiltration),
- narrative cadence (danger → struggle → escape),
- domain-specific motifs (honour, courage, loyalty, peril),
- **triadic narrative structure** (three-part arcs, three-stage conflicts, three-agent scenes).

The last element is particularly notable: GLP frequently produces triadic configurations such as *Marvin–Alice–antagonist*, *pursuit–battle–escape*, or *discovery–danger–resolution*. This triadic geometry emerges naturally from GLP’s grammar cadence and page activation mechanisms, which favour compact, modular narrative units. The continuous lemma embedding space reinforces this behaviour by clustering semantically related lemmas into three-part interaction motifs. The LLM remastering step does not invent this triadic structure; it merely interprets and connects the GLP-generated fragments, preserving the underlying world model.

It is worth noting that the training domain itself exhibits a pronounced triadic character. Burroughs’ Barsoom narratives frequently unfold through three-part conflicts, trio-based character groupings, and modular three-stage adventure arcs. GLP absorbs this structural ontology without reproducing the original text, and the hybrid GLP+LLM pipeline preserves the triadic geometry while providing stylistic coherence.

This emergent triadic behaviour is one of the reasons the resulting volume carries the subtitle *Triadic Pulp Fiction*: the stories inhabit a Barsoom-derived world while exhibiting the modular, three-part narrative rhythm characteristic of GLP’s structural generative process.

5.2 LLM remastering as interpretive refinement

Because the remastering prompt is minimal and non-directive, Mistral 7B acts primarily as an interpreter of GLP’s proto-stories. It performs:

- **gap filling**: smoothing transitions between GLP-generated segments,
- **arc formation**: shaping GLP’s episodic fragments into coherent narrative arcs,
- **stylistic elevation**: improving readability, pacing, and emotional clarity,
- **character grounding**: maintaining Marvin and Alice as consistent protagonists.

Crucially, the LLM does not override GLP’s world model. The scenes, conflicts, settings, and motifs remain structurally Barsoomian, even though the surface text is transformer-polished.

5.3 Evidence of world absorption without textual replication

The analysed corpus demonstrates that GLP does not memorise or reproduce the training text. Instead, it reconstructs the latent world model of *Thuvia, Maid of Mars*. The stories contain:

- barroom confrontations,
- palace infiltrations,
- rooftop escapes,
- duels with Pathan warriors,
- alliances with princesses,

— all recognisable Barsoomian motifs — yet none of these scenes appear in the original book. The LLM remastering step preserves this property: it refines GLP’s proto-stories but does not introduce Barsoom text or plotlines.

This confirms that GLP learns a structural ontology rather than a corpus, and that the hybrid pipeline maintains this ontology through refinement.

5.4 Emergent recombination under hybrid generation

The stories exhibit a characteristic form of recombinative creativity. GLP provides new configurations of familiar elements, while the LLM ensures coherence:

- GLP recombines domain motifs into new proto-scenes,
- the LLM interprets these scenes and links them into arcs,
- GLP supplies world fidelity, the LLM supplies narrative fluency.

This hybrid behaviour mirrors human–LLM collaboration: GLP plays the role of a world-builder, and the LLM plays the role of a stylistic editor.

5.5 Implications for hybrid structural generative systems

The corpus demonstrates that:

- GLP provides a stable world engine,
- GLP does not replicate training text,
- GLP generates structurally coherent proto-stories,
- LLM remastering elevates these proto-stories without altering the world model,
- minimal prompting is sufficient for high-quality hybrid output.

This confirms that learnable lemma embeddings strengthen GLP’s ability to reconstruct and extend narrative worlds, and that hybrid GLP+LLM pipelines can produce rich, coherent narratives while preserving structural fidelity to the learned domain.

6 Narrative Behaviour Under a Hyde-Trained GLP World Model

To evaluate whether GLP can internalise and reproduce fundamentally different narrative domains, we trained a second 34 MB GLP model on *Strange Case of Dr Jekyll and Mr Hyde* using the same hyperparameters, curriculum structure, and epoch count as the Barsoom model analysed in the previous section. The narrator pipeline and remastering prompt were kept identical: the LLM received only minimal stylistic guidance and the fixed protagonist names *Marvin* and *Alice*, with no hints, constraints, or references to the Hyde domain. This setup allows us to isolate the contribution of the GLP world model itself.

6.1 GLP as a gothic structural substrate

Across all generated stories, the Hyde-trained GLP model exhibits a distinct and consistent narrative geometry that differs sharply from the Barsoom world model. The stories contain:

- claustrophobic interiors (mines, basements, laboratories, guest houses),
- shadow-filled corridors, echoing chambers, and dimly lit rooms,
- moral ambiguity, paranoia, and introspective tension,
- mysterious strangers, concealed motives, and cryptic letters,
- chemical apparatus, phials, vials, and experimental devices,
- transformations, inner turmoil, and dual-identity motifs,
- foggy streets, deserted avenues, and oppressive atmospheres.

These motifs are canonical to the gothic psychological domain of *Jekyll and Hyde*, yet none of the generated scenes appear in the original text. The GLP world model reconstructs the latent ontology of the domain—its spatial geometry, emotional cadence, and thematic attractors—without memorising or replicating the training corpus.

6.2 Contrast with the Barsoom world model

The contrast with the Barsoom-trained GLP model is striking. Where Barsoom stories exhibit:

- triadic action grammar,
- modular three-part conflicts,
- heroic rescue arcs,
- palace infiltrations and desert escapes,
- stable social ontologies (nobles, warriors, princesses),

the Hyde stories instead display:

- introspective psychological conflict,
- moral uncertainty and internal struggle,
- claustrophobic spatial geometry,
- chemical experimentation and hidden laboratories,
- ambiguous social interactions and unreliable witnesses.

This demonstrates that GLP does not impose a fixed narrative style. Instead, it absorbs the structural ontology of the training domain and reproduces it faithfully under pure sampling, even when the narrator pipeline and remastering prompt contain no domain hints.

6.3 Evidence of world absorption without stylistic prompting

The Hyde corpus shows that GLP can generate:

- gothic introspection without being prompted for gothic style,
- Hyde-like characters without being told to include Hyde,
- laboratory scenes without any mention of science or chemistry,
- foggy London streets without geographical hints,
- moral duality without explicit thematic guidance.

The LLM remastering step again acts only as an interpretive refiner: it smooths transitions and improves readability, but it does not introduce Hyde motifs. All domain-specific behaviour originates from the GLP world model.

6.4 Implications for multi-world GLP models

The Barsoom and Hyde experiments together demonstrate that GLP can internalise multiple, structurally distinct narrative worlds. Each world model produces coherent proto-stories that reflect its learned ontology, and hybrid GLP+LLM pipelines preserve this ontology even under minimal prompting.

This confirms that GLP is not a stylistic variant of transformer models, but a *trainable world engine* capable of reconstructing and extending diverse narrative domains through its hybrid symbolic–geometric architecture.

7 Ablation: Learning Dynamics in the Lemma Embedding Space

This section presents a minimal but informative ablation study designed to isolate the effect of the newly introduced *learnable lemma embeddings* and the associated *damping schedule*. To avoid confounding factors, we use an extremely small curriculum and a very small GLP model of approximately 9 MB. All generations are produced using pure sampling.

The dataset is intentionally tiny: a dark-forest micro-world containing rabbits, raccoons, foxes, caverns, fissures, mist, pulses, and other recurring motifs. This scale allows us to observe the structural behavior of the embedding space under different learning-rate regimes without the smoothing effect of large corpora.

7.1 Experimental Setup

We vary three parameters governing the lemma embedding dynamics:

- the learning rate α ,
- the damping factor d controlling exponential decay of α ,
- the number of transitions before α reaches its minimum.

Five configurations are tested:

1. **No learning:** $\alpha = 0$.
2. **Low learning rate with damping:** $\alpha = 0.1 \rightarrow 0.01$.
3. **Low learning rate without damping:** $\alpha = 0.1$ constant.

4. **High learning rate with damping:** $\alpha = 0.5 \rightarrow 0.05$.
5. **High learning rate with faster damping:** $\alpha = 0.5 \rightarrow 0.05$ with half the transition horizon.

All models are trained for 10k epochs, producing approximately 2.7M training steps.

7.2 Results

The five runs reveal a clear and interpretable pattern:

1. **No learning** ($\alpha = 0$). Lemma embeddings remain at their hash initialization. The model collapses into a small set of attractors (e.g. *cavern–tendrill–surge*), with minimal variation and repetitive structure. This demonstrates that learnable embeddings are structurally necessary for world absorption.
2. **Low learning rate with damping** ($0.1 \rightarrow 0.01$). Embeddings evolve quickly at first and then stabilize. The resulting world is coherent, varied, and preserves the triadic structure of the micro-world. This is the most stable configuration at small scale.
3. **Low learning rate without damping** (0.1 constant). Embeddings continue to drift throughout training. The world remains recognizable but becomes increasingly chaotic, with weaker scene structure and looser lemma associations. This demonstrates the role of damping in preventing semantic drift.
4. **High learning rate with damping** ($0.5 \rightarrow 0.05$). Embeddings absorb the world extremely quickly, producing richer variation and sharper motifs. Damping prevents collapse and yields a stable world after the initial “turbo phase”. This configuration shows that higher learning rates can be beneficial when paired with appropriate damping.
5. **High learning rate with faster damping**. Reducing the transition horizon produces an even cleaner balance between rapid absorption and stable consolidation. The world remains coherent, varied, and free of drift, confirming that the damping schedule is a critical control mechanism.

7.3 Discussion

Across all five conditions, the ablation demonstrates that:

- learnable lemma embeddings are essential for world formation,
- damping is required to prevent drift in the embedding space,
- higher learning rates can accelerate world absorption when properly damped,
- even at extremely small scale, the learned embedding space outperforms fixed-bit encodings such as tri-hot.

These results support the relevance of the proposed feature: the lemma embedding space is not merely a convenience, but a structural component of GLP’s world engine. The ablation also shows that the mechanism behaves predictably across learning-rate regimes, reinforcing the theoretical claims made earlier in the paper.

8 Conclusion

This paper introduced a learnable lemma embedding layer for Grammar Lemma Paged Language Models (GLP), replacing the earlier bit-vector and tri-hot encoding with a continuous, fixed-dimensional geometric representation. The new layer integrates seamlessly into GLP’s dual-channel architecture: grammar remains deterministic and structurally governing, pages remain discrete and domain-isolating, and lemma selection becomes a geometric process driven by cosine-similarity scoring and grammar compatibility.

A key contribution of this work is the α -damped post-batch update rule for lemma embeddings. Empirical tests demonstrate that damping is essential for long-term stability: without damping, extended training causes embedding drift, degraded page coherence, and slower generation; with damping, the embedding space converges smoothly, and sampling remains fast even after millions of training steps. This shows that the learnable embedding mechanism is not merely an optimisation detail but a structural component of GLP’s generative behaviour.

The learnable lemma embedding layer significantly improves practical performance. Sampling is faster, narrative variation is richer, and fewer epochs are required to obtain a usable model. Vocabulary growth becomes fully dynamic, decoupled from model geometry, and hybrid integration with transformer models becomes more natural due to the continuous embedding space.

The approach was validated through a full narrator-pipeline demonstration. A 33 MB GLP model trained on *Thuvia, Maid of Mars* generated coherent pulp-fiction stories using sampling, which were subsequently remastered by transformer models (Mistral 7B and a larger LLM). The resulting volume, *The Cosmic Parrot: Triadic Pulp Fiction*, illustrates how a lightweight structural model equipped with learnable lemma embeddings can serve as a fast, stable, and domain-faithful generative engine within hybrid GLP+LLM systems.

Together, these results position learnable lemma embeddings as a foundational extension of GLP. They preserve the architecture’s structural commitments while unlocking a continuous semantic space that enhances expressiveness, stability, and practical usability. This work therefore strengthens GLP as a modular, interpretable, and scalable alternative to monolithic transformer architectures, and provides a concrete pathway for future research in structurally governed generative systems within the Triadic Cosmos ecosystem.

A Comparison of Embedding Geometry in GLP and LLM

This appendix clarifies the conceptual and architectural differences between the learnable lemma embeddings introduced in GLP and the embedding mechanisms used in transformer-based language models (LLMs). Although GLP now includes a learnable geometric layer, its embedding dynamics, training behaviour, and generative role differ fundamentally from those of LLMs.

A.1 LLM Embeddings: Unified, Fully Learnable, and Backpropagated

Transformer models employ a single, unified embedding space:

- **Input embeddings** are fully learnable. Each vocabulary token is assigned a vector that evolves through gradient descent.
- **Output logits** are computed through a dot-product between the hidden state and an output matrix that is often *tied* to the input embedding matrix.
- **Backpropagation** updates both input and output embeddings jointly, producing a monolithic, high-dimensional space in which grammar, semantics, and token identity coexist.
- **Softmax over the full vocabulary** determines token selection.

Thus, LLMs rely on a single geometric space for both encoding and decoding, and all token behaviour is governed by gradient descent within that space.

A.2 GLP Embeddings: Deterministic Input, Partially Learnable Output

GLP differs structurally:

- **Deterministic input embeddings.** Every GLP token is associated with a small set of fixed, deterministic vectors used for structural encoding. These include grammar-state encodings, page identifiers, narrative-memory projections, and context-compression signals. These vectors do not form a unified embedding space, do not encode semantic similarity, and do not participate in backpropagation. Their role is purely organisational: they provide compact structural descriptors that allow GLP’s dual-channel architecture to maintain long-horizon coherence without learned input embeddings.
- **Output embeddings are learnable.** Each lemma token ℓ has a stored embedding e_ℓ used exclusively for *output selection* via cosine similarity. These embeddings are deterministically initialised (via hashing) and become partially learnable during training.
- **No backpropagation into lemma embeddings.** The lemma MLP is trained through gradient descent, but the stored lemma embeddings are updated through an α -damped mixing rule:

$$e_\ell \leftarrow (1 - \alpha)e_\ell + \alpha\hat{e},$$

where $\hat{e}$ is the predicted embedding. This update is not a gradient step but a controlled geometric interpolation, allowing rapid early learning and smooth long-term convergence.

- **Page-restricted cosine ranking.** Lemma selection is performed by computing

$$\cos(\hat{e}, e_\ell)$$

for candidates within the active page, filtered by grammar compatibility. GLP does not perform a full-vocabulary softmax.

GLP therefore separates structure and geometry: grammar and context remain symbolic and deterministic, while lemma selection becomes geometric and continuous.

A.3 Structural Consequences

These differences have several implications:

- **GLP is not a monolithic embedding model.** Structural input signals are deterministic and non-learnable; only lemma output embeddings participate in learning.
- **GLP’s embedding space is an output space only.** It represents *world geometry* (motifs, clusters, attractors), not input token identity or syntactic encoding.
- **GLP’s embedding updates are stable and interpretable.** The α -damped mixing rule avoids the instability associated with gradient descent on discrete vocabularies.
- **GLP’s generative behaviour remains structurally governed.** Cosine ranking operates within grammar-filtered pages, preserving canonical structure even as semantic variation increases.

A.4 Summary

Although GLP now includes a learnable geometric layer, it remains fundamentally different from transformer models:

- LLMs learn *input* embeddings; GLP uses deterministic structural embeddings for all input signals.
- LLM embeddings are updated by backpropagation; GLP lemma embeddings are updated by α -blended interpolation.
- LLMs use a global softmax over the entire vocabulary; GLP applies a local softmax only over the top- k cosine-ranked lemma candidates within the active page.
- LLMs unify grammar and semantics in one space; GLP keeps grammar symbolic and deterministic while adding a geometric lemma layer.

This distinction is essential: GLP’s learnable lemma embeddings enhance semantic expressiveness without altering its structural commitments, preserving GLP as a hybrid symbolic-geometric world engine rather than a transformer-style language model.

B Triadic Cosmos Ecosystem Context

The present paper is part of the broader *Triadic Cosmos* ecosystem: a unified research program that develops a compositional ontology of intelligence together with modular, inspectable computational architectures. The ecosystem spans conceptual theory, executable reference implementations, and fully documented empirical demonstrations, forming a coherent framework for studying hybrid intelligent systems.

All materials, extended notes, and evolving documentation are publicly available in the open GitHub project:

<https://github.com/triadic-cosmos>

Within the `triadic-system` repository, a complete reference implementation of the GLP engine, the sequential narrator pipeline, and all supporting modules is provided under the AGPL-3.0 license. These components form a modular stack: deterministic grammar structure, paged lemma vocabularies, learnable lemma embeddings, PRE-based sentence validation, and hybrid LLM remastering.

The learnable lemma embedding layer introduced in this paper integrates directly into this ecosystem. It is used in the narrator pipeline to generate long-form narrative content, followed by transformer-based remastering. A 33 MB GLP model trained on *Thuvia, Maid of Mars* (Gutenberg) serves as the structural engine for a new volume of pulp-fiction stories titled *The Cosmic Parrot: Triadic Pulp Fiction*. This book demonstrates the practical effectiveness of the GLP+LLM hybrid pipeline: GLP provides canonical structure, domain fidelity, and narrative stability, while transformer models (Mistral 7B and a larger LLM) provide stylistic refinement and surface fluency.

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